



Cambridge IGCSE™

BIOLOGY

0610/52

Paper 5 Practical Test

February/March 2026

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the February/March 2026 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **14** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	correct point or mark awarded
	incorrect point or mark not awarded
	information missing or insufficient for credit
	allow or accept
	incorrect or insufficient point ignored while marking the rest of the response
	contradiction in response, mark not awarded
	benefit of the doubt given
	error carried forward applied
	point has been noted, but no credit has been given or blank page seen

Annotation	Meaning
	correct awarding one mark from marking point or marking group 1. similar numbered ticks are used for marking point or marking groups 2, 3, 4 etc.
	pages are linked together
	used to highlight parts of an extended response
	used to highlight parts of an extended response
	Point already given
	Maximum mark reached
	Key point attempted / working towards marking point / incomplete answer / response seen but not credited / blank page seen
	Maximum number of marks for a marking point has been awarded.

Mark Scheme Abbreviations:	
;	separates marking points
/	alternative responses for the same marking point
R	reject the response
A	accept the response
I	ignore the response
ecf	error carried forward
AVP	any valid point
ora	or reverse argument
AW	alternative wording
<u>underline</u>	actual word given must be used by candidate (grammatical variants excepted)
()	the word / phrase in brackets is not required but sets the context
max	indicates the maximum number of marks that can be given
MP	marking point

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Question	Answer	Marks	Guidance												
1(a)(i)	12 ;;	2	MP1 calculation of surface area of one cube: $0.5 \times 0.5 \times 6$ 1.5 MP2 surface area multiplied by 8:												
1(a)(ii)	1 table drawn with minimum two columns and a header line ; 2 appropriate column headings for independent and dependent variables ; 3 any three values for their IV <u>and</u> any three recorded heights ; 4 the height of foam in test tube 8 being larger than test tube 1 ; <table border="1"> <tr> <td colspan="2">independent variable: OR</td><td colspan="2">dependent variable: OR</td></tr> <tr> <td>number of pieces</td><td>(total surface) area in cm²</td><td>height / distance / length (of foam) in mm</td><td>height / distance / length (of foam) in cm</td></tr> <tr> <td>any value any value any value</td><td>any value any value any value</td><td></td><td></td></tr> </table>	independent variable: OR		dependent variable: OR		number of pieces	(total surface) area in cm ²	height / distance / length (of foam) in mm	height / distance / length (of foam) in cm	any value any value any value	any value any value any value			4	
independent variable: OR		dependent variable: OR													
number of pieces	(total surface) area in cm ²	height / distance / length (of foam) in mm	height / distance / length (of foam) in cm												
any value any value any value	any value any value any value														
1(a)(iii)	higher / faster / more, enzyme / catalase (activity), when <u>surface area</u> (of potato cubes) is larger ; ora	1													

Question	Answer	Marks	Guidance
1(a)(iv)	<i>any two from:</i> volume / 10 cm ³ , of hydrogen peroxide ; mass / weight / volume (total) of 1 cm ³ potato ; concentration of hydrogen peroxide ; diameter of test-tubes ; same tissue ;	2	
1(a)(v)	hydrogen peroxide / cutting (potato) / (use of) scalpel / blade / knife ;	1	

Question	Answer	Marks	Guidance
1(a)(vi)	<i>mark in pairs</i> 1 <i>error</i> measuring foam e.g. top of foam is uneven / difficult to decide where to measure to / bubbles might pop ; 2 <i>improvement</i> measure the volume, of gas collected ; 3 <i>error</i> potato pieces were left in hydrogen peroxide for different lengths of time / height of foam was not measured after the same length of time (in hydrogen peroxide) ; 4 <i>improvement</i> do the experiments one at a time / use a staggered start / add all the pieces at the same time ; 5 <i>error</i> temperature is not controlled ; 6 <i>improvement</i> use of thermostatically controlled water-bath ; 7 <i>error</i> pH is not controlled ; 8 <i>improvement</i> use of a (pH) buffer ; 9 <i>error</i> cubes could be stuck together / not the same surface area ; 10 <i>improvement</i> stir the cubes if stuck ;	4	Read through (no list rule) Accept errors qualified in improvements

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Question	Answer	Marks	Guidance
1(a)(vi)	11 <i>error</i> cutting the cubes to an accurate size ; 12 <i>improvement</i> use a borer / AW / description / machine <u>and</u> cut with ruler ; 13 <i>error</i> AVP ; 14 <i>improvement</i> AVP ;		
1(b)	blue-black ;	1	
1(c)(i)	<i>dependent variable</i> time taken (for potato / discs) to rise / reach surface ; <i>independent variable</i> <u>concentration</u> of copper sulfate ;	2	
1(c)(ii)	boiled / cooked / AW, potato (pieces) / AW ; replace hydrogen peroxide with water ;	1	
1(c)(iii)	Axes – labelled with units ; Scale – suitable linear scale and data occupies at least half the grid in both directions ; Plot – 5 points plotted accurately $\pm$ half a small square ; Line – suitable line drawn ;	4	x-axis – conc(entr)ation of copper sulfate in mol per dm³ y-axis – time (for disc of potato to rise) in s(econds)

Question	Answer	Marks	Guidance
1(c)(iv)	intercept indicated on candidate's graph at $80 \text{ s} \pm \frac{1}{2}$ a small square ; concentration correct for intercept at $80 \text{ s} \pm \frac{1}{2}$ a small square ;	2	expected answer 0.006 OR 6×10^{-3} (mol per dm^3) MP2 Allow ecf from MP1 or missing indication on graph

Question	Answer	Marks	Guidance
2(a)(i)	Outline - single clear line <u>and</u> no shading ; Size – drawing is at least 74 mm wide ; Detail ;;	4	Detail 1 exactly 5 seeds on lower side with 2nd <u>and</u> 3rd seeds touching each other <u>and</u> at least one of that pair is touching the septum (note: this could be lower part of septum) Detail 2 line across base of fruit to show where the stalk attaches (stalk does not need to be shown) <u>and</u> fruit wall shown as a double line
2(a)(ii)	(PQ =) 74 ± 1 mm ; 43.5 ;;	3	MP1 measuring line MP2 correct calculation of actual size: $74(\pm 1) \div 1.7$ 43.5294.... etc. MP3 correctly rounded to three significant figures ecf for MP2 and 3 if MP1 is wrongly stated in cm with no unit change ecf from previous step for incorrect numbers MP3 ecf from reversed numerator and denominator
2(b)(i)	(brick) red / orange / yellow / green (precipitate) ;	1	
2(b)(ii)	84.18 ;;	2	MP1 correct calculation: $\frac{36.1 - 19.6}{19.6} \times 100$ 84.183.... MP2 correct rounding to two d.p.

Question	Answer	Marks	Guidance
3	1 <i>independent variable</i> ; two different wind speeds 2 <i>dependent variable</i> ;; • measure mass of, leaves / plant, before and after • measuring the movement of bubble / meniscus, in capillary tube of potometer • measuring mass of pot and soil before and after • measuring, mass / volume, of (shoot / leaf and) water / dye, before and after • observing change in colour of (anhydrous) cobalt chloride paper • distance dye has moved up a, plant / stalk 3 wait for fixed period of time / record time after a set period of change ; <i>detail of method</i> ;; 4 method for obtaining / maintaining, two wind speeds 5 description of, suspending leaves (on a string) / use of potometer / counting number of sections cut with dye present 6,7 <i>constant variables</i> ;; • temperature • humidity • light intensity • carbon dioxide concentration • size / surface area / number, of leaves • type / species / variety / age / condition, of, plant / leaves • same dye (if using dye) 8 investigation carried out at least 3 times / repeat 2 more times (for each wind speed) ; 9 relevant description of and name appropriate safety equipment ;	6	